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Fall 2026

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- oscilloscope
- pre-amplifier
- amplifier
- single channel analyzer
- counter/timer
- multi channel analyzer

2.1.2 Measurement Setup

- Split the pre-amp output and connect the one branch to the scope and the other to the amplifier
- Connect the pre-amplifier to the high voltage power supply and the G-M tube
- Connect the pre-amp to the amplifier, the amplifier to the SCA, and finally the SCA to the counter/timer
- Set the SCA to integral mode

2.2 Counting Curve Measurements

2.2.1 Determining the LLD

When creating a counting curve, it is important to set the LLD such that noise is just discriminated at the plateau.

- Set the high voltage to 800 V
- Remove the source from the detector chamber
- Find the LLD value that is just large enough to discriminate the noise pulses (should be a small value, less than 1 V). This requires monitoring the counts on the counter/timer. If this value is below 0.1 V, just set the LLD to 0.1 V.

2.2.2 Generation of the counting curve

- Return the source to the detector chamber
- Reduce the high voltage until the counter-timer records almost no pulses
- Now increase the high voltage in increments of 10-30 V and measure the **count rate** as well as the **average pulse height** from the pre-amp. You will notice a “knee” and a “plateau” of the counting rate curve. Take measurements for at least 10 different voltages. Wait for ~30 s for the voltage to stablize every time you change the voltage.
- DO NOT EXCEED 1000 V. REDUCE BIAS AT FIRST SIGN OF ABRUPT COUNT RATE INCREASE.

3 Pulse Height Consistency

The purpose of this section is to establish the correlation between pulse height and incident source particle and energy.

- Select the ideal operating voltage from your counting curve. You want to select a point on the higher voltage side of the smoothest region in your plateau.
- Connect the amplifer to MCA and use Maestro software to visualize the spectrum.
- For 3 beta-emitter sources (one at a time), record a spectrum for ~30s. Record the pulse amplitude centroid, and record the gross counts in the spectrum.

4 Beta Attenuation

The purpose of this section is to use our G-M measurements to estimate the absorption coefficient for beta particles. For this part, you can use either SCA or MCA.

- Keep your detector at the ideal operating voltage, as determined from your counting curve.
- Choose a beta source with sufficient activity to give you at least several thousand counts per second on the second tray position (make sure this source is in Figure 2.16 of Knoll).
- Record the counts with no source in the chamber (background)
- Record the counts with 8-9 different aluminum absorber thicknesses → Make sure to count longer with thicker absorbers in order to maintain a small fractional standard deviation for each measurement.

5 Dead Time and Recovery Time

The goal is to measure the dead time of our system using two independent techniques.

5.1 Oscilloscope Dead Time Measurement

If we don't let the screen refresh between triggers, we will accumulate a bunch of pulses on the display. Sometimes a pulse will come soon after the initial pulse. After accumulating enough pulses we can measure the minimum distance between two pulses.

- Keep your detector at the ideal operating voltage, as determined from your counting curve.